

Marlene's grid + LMM — primary

Three comparisons $\times$ five specifications (permutation OLS, 1 df) plus the mixed-model omnibus on the full category factor. All measures oriented so higher = more selective / more distinct. beta positive = the first-named category set is relatively LESS affected in group B.

peak_z (univariate peak selectivity)

1. A = LH ctrl B = LH-intact pt — nA=36, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	+1.372	—	0.0298 *
binB word face+house+object	+1.537	—	0.0474 *
binC word+face+house object	+1.560	—	0.0312 *
cont LI z>1.96	+7.595	—	0.0086 **
cont LI z>2.33	+6.992	—	0.0098 **
LMM omnibus category $\times$ group	—	5.57 (3)	0.1347

LMM covariates: age β =+0.149 p=0.0600

2. A = RH ctrl B = RH-intact pt — nA=36, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	+1.264	—	0.1194
binB word face+house+object	+0.749	—	0.3611
binC word+face+house object	+0.348	—	0.6789
cont LI z>1.96	+5.450	—	0.1152
cont LI z>2.33	+4.997	—	0.1176
LMM omnibus category $\times$ group	—	3.27 (3)	0.3522

LMM covariates: age β =+0.306 p=0.0002 **sig**

3. A = LH-intact pt B = RH-intact pt — nA=12, nB=12 (*patient-group $\times$ category; no control reference*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.134	—	0.8902
binB word face+house+object	-1.394	—	0.1888
binC word+face+house object	+0.143	—	0.8820
cont LI z>1.96	-2.792	—	0.5227
cont LI z>2.33	-2.905	—	0.4723
LMM omnibus category $\times$ group	—	3.34 (3)	0.3426

LMM covariates: age β =+0.227 p=0.1114; surgery (OTC vs Hemi) β =+2.179 p=0.0209 **sig**

RSA distinctiveness

1. A = LH ctrl B = LH-intact pt — nA=35, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.213	—	0.0616
binB word face+house+object	+0.051	—	0.7107
binC word+face+house object	+0.115	—	0.4387
cont LI z>1.96	+0.747	—	0.2032
cont LI z>2.33	+0.662	—	0.2304
LMM omnibus category $\times$ group	—	4.03 (3)	0.2585

LMM covariates: age β =+0.004 p=0.5594

2. A = RH ctrl B = RH-intact pt — nA=35, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	-0.158	—	0.2080
binB word face+house+object	-0.214	—	0.1270
binC word+face+house object	-0.196	—	0.2246
cont LI z>1.96	-0.959	—	0.1356
cont LI z>2.33	-0.889	—	0.1328
LMM omnibus category $\times$ group	—	3.21 (3)	0.3608

LMM covariates: age β =-0.003 p=0.7063

3. A = LH-intact pt B = RH-intact pt — nA=12, nB=12 (*patient-group $\times$ category; no control reference*)

specification	beta	χ^2 (df)	p
binA word+face house+object	-0.056	—	0.6995
binB word face+house+object	+0.029	—	0.8580
binC word+face+house object	+0.021	—	0.9190
cont LI z>1.96	-0.099	—	0.8938
cont LI z>2.33	-0.081	—	0.9056
LMM omnibus category $\times$ group	—	0.49 (3)	0.9212

LMM covariates: age β =-0.006 p=0.5987; surgery (OTC vs Hemi) β =-0.025 p=0.7628

between-category pattern similarity (6 pairs)

1. A = LH ctrl B = LH-intact pt — nA=35, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.106	—	0.3541
binB word face+house+object	-0.085	—	0.4405
binC word+face+house object	-0.046	—	0.6357
cont LI z>1.96	+0.117	—	0.8234
cont LI z>2.33	+0.083	—	0.8660
LMM omnibus category $\times$ group	—	7.93 (5)	0.1604

LMM covariates: age β =+0.001 p=0.8409

2. A = RH ctrl B = RH-intact pt — nA=35, nB=12

specification	beta	χ^2 (df)	p
binA word+face house+object	-0.136	—	0.1604
binB word face+house+object	-0.498	—	0.0016 **
binC word+face+house object	-0.078	—	0.2655
cont LI z>1.96	-1.544	—	0.0074 **
cont LI z>2.33	-1.505	—	0.0062 **
LMM omnibus category $\times$ group	—	21.81 (5)	0.0006 ***

LMM covariates: age β =-0.007 p=0.3274

3. A = LH-intact pt B = RH-intact pt — nA=12, nB=12 (*patient-group $\times$ category; no control reference*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.004	—	0.9712
binB word face+house+object	+0.060	—	0.7021
binC word+face+house object	+0.152	—	0.2705
cont LI z>1.96	+0.152	—	0.8030
cont LI z>2.33	+0.136	—	0.8170
LMM omnibus category $\times$ group	—	2.07 (5)	0.8391

LMM covariates: age $\beta=-0.011$ $p=0.3557$; surgery (OTC vs Hemi) $\beta=-0.065$ $p=0.4140$

Supplemental — crossed and paired

Comparisons 4 and 5 pit a patient's intact hemisphere against the OPPOSITE control hemisphere, so they confound resection with normal hemispheric asymmetry. Comparison 6 measures that asymmetry alone: the same controls, both hemispheres, paired within subject. Read 4 and 5 against 6.

peak_z (univariate peak selectivity)

4. A = RH ctrl B = LH-intact pt [crossed] — nA=36, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+1.131	—	0.1462
binB word face+house+object	+2.143	—	0.0044 **
binC word+face+house object	+0.205	—	0.8038
cont LI z>1.96	+8.242	—	0.0094 **
cont LI z>2.33	+7.902	—	0.0064 **
LMM omnibus category $\times$ group	—	6.72 (3)	0.0816

LMM covariates: age $\beta=+0.221$ $p=0.0099$ **sig**

5. A = LH ctrl B = RH-intact pt [crossed] — nA=36, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+1.506	—	0.0270 *
binB word face+house+object	+0.143	—	0.8584
binC word+face+house object	+1.703	—	0.0252 *
cont LI z>1.96	+4.803	—	0.1350
cont LI z>2.33	+4.088	—	0.1754
LMM omnibus category $\times$ group	—	7.12 (3)	0.0683

LMM covariates: age $\beta=+0.233$ $p=0.0027$ **sig**

6. A = LH ctrl B = RH ctrl [paired] — nA=36, nB=36 (*within subject*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.193	—	0.5517
binB word face+house+object	-0.383	—	0.0834
binC word+face+house object	+0.411	—	0.0998
cont LI z>1.96	+0.157	—	0.8448
cont LI z>2.33	+0.088	—	0.9000
LMM omnibus category $\times$ group	—	8.94 (3)	0.0301 *

LMM covariates: age $\beta=+0.209$ $p=0.0117$ **sig**

RSA distinctiveness

4. A = RH ctrl B = LH-intact pt [crossed] — nA=35, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	-0.102	—	0.4009
binB word face+house+object	-0.243	—	0.0914
binC word+face+house object	-0.217	—	0.1904
cont LI z>1.96	-0.861	—	0.1692
cont LI z>2.33	-0.809	—	0.1648
LMM omnibus category $\times$ group	—	4.20 (3)	0.2410

LMM covariates: age β = -0.004 p=0.6488

5. A = LH ctrl B = RH-intact pt [crossed] — nA=35, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.157	—	0.1686
binB word face+house+object	+0.080	—	0.5475
binC word+face+house object	+0.136	—	0.3251
cont LI z>1.96	+0.648	—	0.2631
cont LI z>2.33	+0.581	—	0.2821
LMM omnibus category $\times$ group	—	2.21 (3)	0.5292

LMM covariates: age β = +0.005 p=0.4974

6. A = LH ctrl B = RH ctrl [paired] — nA=35, nB=35 (*within subject*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.234	—	0.0004 ***
binB word face+house+object	+0.297	—	0.0016 **
binC word+face+house object	+0.160	—	0.0008 ***
cont LI z>1.96	+0.613	—	0.0014 **
cont LI z>2.33	+0.560	—	0.0018 **
LMM omnibus category $\times$ group Δ no convergence	—	18.36 (3)	0.0004 ***

LMM covariates: age β = +0.003 p=0.6957

between-category pattern similarity (6 pairs)

4. A = RH ctrl B = LH-intact pt [crossed] — nA=35, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	-0.140	—	0.1338
binB word face+house+object	-0.559	—	0.0002 ***
binC word+face+house object	-0.230	—	0.0144 *
cont LI z>1.96	-1.696	—	0.0030 **
cont LI z>2.33	-1.641	—	0.0028 **
LMM omnibus category $\times$ group	—	26.34 (5)	7.7e-05 ***

LMM covariates: age β = -0.006 p=0.4264

5. A = LH ctrl B = RH-intact pt [crossed] — nA=35, nB=12 (*crossed; confounded with normal asymmetry*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.110	—	0.3481
binB word face+house+object	-0.025	—	0.8314
binC word+face+house object	+0.106	—	0.1606
cont LI z>1.96	+0.270	—	0.6347
cont LI z>2.33	+0.218	—	0.6771
LMM omnibus category × group	—	5.59 (5)	0.3483

LMM covariates: age β =+0.000 p=0.9778

6. A = LH ctrl B = RH ctrl [paired] — nA=35, nB=35 (*within subject*)

specification	beta	χ^2 (df)	p
binA word+face house+object	+0.151	—	0.0134 *
binB word face+house+object	+0.356	—	0.0002 ***
binC word+face+house object	+0.090	—	0.0554
cont LI z>1.96	+0.403	—	0.0276 *
cont LI z>2.33	+0.374	—	0.0270 *
LMM omnibus category × group	—	30.69 (5)	1.1e-05 ***

LMM covariates: age β =-0.000 p=0.9563

TFCE has no subject × category value, so it cannot take this form. Report the existing cluster table instead.

* p<.05 ** p<.01 *** p<.001